

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- Let $P(x) = kx^3 + 2k^2x^2 + k^3$. The sum of all real numbers k for which $(x - 2)$ is a factor of $P(x)$, is
(A) 4 (B) 8 (C) -4 (D) -8
MFN001
- Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1; P(2) = 2; P(3) = 3; P(4) = 4; P(5) = 5$ and $P(6) = 6$, then find the value of $P(7)$.
(A) 700 (B) 727 (C) 650 (D) 725
MFN002
- The domain of definition of the function, $y(x)$ is given by the equation, $2^x + 2^y = 2$ is -
(A) $0 < x \leq 1$ (B) $0 \leq x \leq 1$ (C) $-\infty < x \leq 0$ (D) $-\infty < x < 1$
MFN003
- Range of function $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$ is given by
(A) $(0, \infty)$ (B) $\left[\frac{1}{2}, 1\right]$ (C) $[1, 2]$ (D) $\left[\frac{1}{4}, 1\right]$
MFN004
- Range of $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}; x \in \left(0, \frac{\pi}{2}\right)$ is -
(A) $(0, 1)$ (B) $(1, \infty)$ (C) $(-1, 0)$ (D) $(-\infty, -1)$
MFN005
- Let $f(x) = \frac{x}{1-x}$ and let α be a real number. If $x_0 = \alpha, x_1 = f(x_0), x_2 = f(x_1), \dots$ & $x_{2011} = -\frac{1}{2012}$ then the value of α is -
(A) $\frac{2011}{2012}$ (B) 1 (C) 2011 (D) -1
MFN006
- If $f_1(x) = 2^{f_2(x)}$, where $f_2(x) = 2012^{f_3(x)}$, where $f_3(x) = \left(\frac{1}{2013}\right)^{f_4(x)}$, where $f_4(x) = \log_{2013} \log_x 2012$, then the range of $f_1(x)$ is -
(A) $(2, \infty)$ (B) $(2012, \infty)$ (C) $(0, \infty)$ (D) $(-\infty, \infty)$
MFN007
- Let $f: \mathbb{R} - \left\{\frac{-15}{2}\right\} \rightarrow \mathbb{R} - \left\{\frac{1}{2}\right\}$ be defined by $f(x) = \frac{x+10}{2x+15}$ then $f(x)$ is -
(A) one-one but not onto (B) many one but not-onto
(C) one-one and onto (D) many one and onto
MFN008
- $f: \mathbb{R} \rightarrow \mathbb{R} f(x) = \frac{2x^2 - 5x + 3}{8x^2 + 9x + 11}$, then f is -
(A) one-one onto (B) many-one onto (C) one-one into (D) many one into
MFN009

10. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} x+3 & , \quad x \in \text{rational} \\ 4x & , \quad x \in \text{irrational} \end{cases}$,

$$g(x) = \begin{cases} x + \sqrt{5} & , \quad x \in \text{irrational} \\ -x & , \quad x \in \text{rational} \end{cases} \text{ then } (f - g)(x) \text{ is -}$$

- (A) one-one & onto (B) neither one-one nor onto
(C) one-one but not onto (D) onto but not one-one

MFN010

11. Let $f: X \rightarrow Y$ be a function such that $f(x) = \sqrt{x-2} + \sqrt{4-x}$, then the set of X and Y for which $f(x)$ is both injective as well as surjective, is -

- (A) $[2,4]$ and $[\sqrt{2}, 2]$ (B) $[3,4]$ and $[\sqrt{2}, 2]$
(C) $[2,4]$ and $[1,2]$ (D) $[2,3]$ and $[1,2]$

MFN011

12. If $f: \mathbb{R} \rightarrow \mathbb{R}$ & $f(x) = \frac{\sin([x]\pi)}{x^2 + 2x + 3} + 2x - 1 + \sqrt{x(x-1)} + \frac{1}{4}$ (where $[x]$ denotes integral part of x), then

$f(x)$ is -

- (A) one-one but not onto (B) one-one & onto
(C) onto but not one-one (D) neither one-one nor onto

MFN012

13. If the function $f: \left[\frac{7}{2}, \infty\right) \rightarrow \left[-\frac{9}{4}, \infty\right)$ be defined by $f(x) = x^2 - 7x + 10$ then $f^{-1}(x)$ is-

- (A) $\frac{7 + \sqrt{9 + 4x}}{2}$ (B) $\frac{7 \pm \sqrt{9 + 4x}}{2}$ (C) $\frac{7 + \sqrt{9 - 4x}}{2}$ (D) $\frac{7 \pm \sqrt{9 - 4x}}{2}$

MFN013

14. $f(x) = [x - 1] + \{x\}^{[x]}$, $x \in (1, 3)$, then $f^{-1}(x)$ is -

(where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function)

- (A) $\begin{cases} x+1 & x \in (1, 2) \\ 2 + \sqrt{x-1} & x \in [2, 3) \end{cases}$ (B) $\begin{cases} x-1 & x \in (1, 2) \\ 2 - \sqrt{x-1} & x \in [2, 3) \end{cases}$
(C) $\begin{cases} x-1 & x \in (0, 1) \\ 2 - \sqrt{x-1} & x \in [1, 2) \end{cases}$ (D) $\begin{cases} x+1 & x \in (0, 1) \\ 2 + \sqrt{x-1} & x \in [1, 2) \end{cases}$

MFN014

15. Let $f: (0, \infty) \rightarrow (1, \infty)$ be a function such that $f(x) = 1 + \frac{3}{2}\sqrt{x}$ and $g(x)$ is inverse of $f(x)$ then the point where $f(x)$ and $g(x)$ intersect is

- (A) $\left(\frac{1}{4}, \frac{1}{4}\right)$ (B) $(4, 2)$ (C) $(4, 4)$ (D) Does not exist

MFN015

16. Which of the following functions is an odd function ?

(A) $|x - 2| + (x + 2) \operatorname{sgn}(x + 2)$ (B) $\frac{1}{x(e^x - 1)} + \frac{1}{2x}$

(C) $\log(\sin x + \sqrt{1 + \sin^2 x})$ (D) $e^{-4x}(e^{2x} - 1)^4$

(where $\operatorname{sgn}(x)$ denotes signum function of x)

MFN016

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function such that $f(10 + x) = f(10 - x) \forall x \in \mathbb{R}$

and $f(20 + x) = -f(20 - x) \forall x \in \mathbb{R}$. Then which of the following statements is true -

- (A) $f(x)$ is odd and periodic (B) $f(x)$ is odd and aperiodic
(C) $f(x)$ is even and periodic (D) $f(x)$ is even and aperiodic

MFN017

18. Period of $f(x) = \{x\} + \left\{x + \frac{1}{3}\right\} + \left\{x + \frac{2}{3}\right\}$ is equal to (where $\{.\}$ denotes fractional part function)

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

MFN018

19. Period of function $f(x) = \min\{\sin x, |x|\} + \frac{x}{\pi} - \left[\frac{x}{\pi}\right]$ (where $[.]$ denotes greatest integer function) is -

(A) $\pi/2$ (B) π (C) 2π (D) 4π

MFN019

20. Let $f(x) = 2x - \left\{\frac{x}{\pi}\right\}$ and $g(x) = \cos x$, where $\{.\}$ denotes fractional part function, then period of $g \circ f(x)$ is -

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{2}$ (D) $\frac{\pi}{4}$

MFN020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let $f(x) = \begin{cases} x^2 - 3x + 4 & ; x < 3 \\ x + 7 & ; x \geq 3 \end{cases}$ and $g(x) = \begin{cases} x + 6 & ; x < 4 \\ x^2 + x + 2 & ; x \geq 4 \end{cases}$, then which of the following is/are true -

(A) $(f + g)(1) = 9$ (B) $(f - g)(3.5) = 1$ (C) $(fg)(0) = 24$ (D) $\left(\frac{f}{g}\right)(5) = \frac{8}{3}$

MFN060

2. Let $f(x) = x^2 + 3x + 2$, then number of solutions to -

(A) $f(|x|) = 2$ is 1 (B) $f(|x|) = 2$ is 3
(C) $|f(x)| = 0.125$ is 4 (D) $|f(|x|)| = 0.125$ is 8

MFN061

3. For each real x , let $f(x) = \max\{x, x^2, x^3, x^4\}$, then $f(x)$ is -

(A) x^4 for $x \leq -1$ (B) x^2 for $-1 < x \leq 0$ (C) $f\left(\frac{1}{2}\right) = \frac{1}{2}$ (D) $f\left(\frac{1}{2}\right) = \frac{1}{4}$

MFN062

4. For the function $f(x) = |x + 3| - |x + 1| - |x - 1| + |x - 3|$, identify correct option(s)

(A) Range of $f(x)$ is $(-\infty, 4]$ (B) maximum value of $f(x)$ is 4
(C) $f(x) = 4$ has infinite solutions (D) $f(x) = 0$ has infinite solutions

MFN063

5. Which of the following statement(s) is(are) correct ?

- (A) If f is a one-one mapping from set A to A , then f is onto.
(B) If f is an onto mapping from set A to A , then f is one-one
(C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that gof is injective, then f must be injective.
(D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

MFN064

6. Consider a function $f(x) = \frac{|x+1|}{x+1} + \frac{x-1}{|x-1|}$, then choose correct option(s) regarding function f

- (A) many one function (B) odd function
(C) bounded function (D) range contain exactly three integers

MFN065

7. Let $f(x) = \begin{cases} x^2 & ; 0 < x < 2 \\ 2x - 3 & ; 2 \leq x < 3 \\ x + 2 & ; x \geq 3 \end{cases}$ Then :-

(A) $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\} = f\left(\frac{3}{2}\right)$ (B) $1 + f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\} = f\left(\frac{5}{2}\right)$
(C) $f\{f(1)\} = f(1) = 1$ (D) $f\{f(1)\} = 2$

MFN066

8. If $\{x\} = \frac{2}{3}$ & $\left[x + \left\{ x + \left[x + \{x + \dots 100 \text{ times} \} \right] \right\} \right] = 5$, then -
 (A) $x = \frac{14}{3}$ (B) $[x] = 5$ (C) $x = \frac{17}{3}$ (D) $[x] = 4$
 (where $[.]$ & $\{.\}$ denotes greatest integer function & fractional part function respectively) **MFN067**
9. Let $f: \{1,2,3,4,5\} \rightarrow \{1,2,3,4,5\}$ is such that $f(x)$ is a one-one function satisfying following condition $f(x) = x + 1$ if and only if x is even (i.e. $f(3) \neq 4$, $f(4) = 5$ etc). Then $f^{-1}(2)$ can be -
 (A) 1 (B) 3 (C) 5 (D) 2 **MFN068**
10. Let $f: [1, \infty) \rightarrow [3, \infty)$ be defined as $f(x) = (\log_2 x)^2 + 2(\log_2 x) + 3$. If $f^{-1}(x)$ is inverse of $f(x)$, then
 (A) $f^{-1}(x) = 2^{-1+\sqrt{x-2}}$ (B) $f^{-1}(x) = 2^{-1-\sqrt{x-2}}$
 (C) domain of $f^{-1}(x)$ is $[2, \infty)$ (D) domain of $f^{-1}(x)$ is $[3, \infty)$ **MFN069**
11. If $f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) = f(x) \quad \forall x \in \mathbb{R}$, then period of function $f(x)$ is greater than -
 (A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 3 **MFN070**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 12 to 13

$$\text{Let } f(x) = \begin{cases} x & ; x < 0 \\ 1-x & ; x \geq 0 \end{cases} \quad \& \quad g(x) = \begin{cases} x^2 & ; x < -1 \\ 2x+3 & ; -1 \leq x \leq 1 \\ x & ; x > 1 \end{cases}$$

On the basis of above information, answer the following questions :

12. Range of $f(x)$ is -
 (A) $(-\infty, 1]$ (B) $(-\infty, \infty)$ (C) $(-\infty, 0]$ (D) $(-\infty, 2]$ **MFN071**
13. Range of $g(f(x))$ is -
 (A) $(-\infty, \infty)$ (B) $[1, 3) \cup (3, \infty)$ (C) $[1, \infty)$ (D) $[0, \infty)$ **MFN072**

Paragraph for Question No. 14 to 15

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying $f(2-x) = f(2+x)$ and $f(20-x) = f(x)$, $\forall x \in \mathbb{R}$.

On the basis of above information, answer the following questions :

14. If $f(0) = 5$, then minimum possible number of values of x satisfying $f(x) = 5$, for $x \in [0, 170]$ is -
 (A) 22 (B) 23 (C) 24 (D) 25 **MFN073**
15. Graph of $y = f(x)$ is symmetrical about $x = k$. Find k -
 (A) 15 (B) 16 (C) 17 (D) 18 **MFN074**

Paragraph for Question No. 16 to 17

Let $f: (-\infty, -2] \rightarrow [3, \infty)$ defined by $f(x) = x^2 + 2x + 3$.

On the basis of above information, answer the following questions :

16. $f^{-1}(x)$ is equal to -
 (A) $\sqrt{x+2} - 1$ (B) $\sqrt{x-2} - 1$ (C) $-\sqrt{x+2} - 1$ (D) $-\sqrt{x-2} - 1$
17. Range of the function $g(x) = f(2x)$ is equal to -
 (A) $[1, \infty)$ (B) $[2, \infty)$ (C) $[3, \infty)$ (D) $[4, \infty)$

MFN075

MFN076

MATCHING LIST TYPE QUESTION

- | 18. | List-I | | List-II |
|-------|--|-----|---|
| | $f(x)$ | | Range |
| (I) | $\frac{\cos^2 x + \cos x + 2}{\cos^2 x + \cos x + 1}$ | (P) | $\left(0, \frac{7}{3}\right]$ |
| (II) | $\left \frac{(\sqrt{\cos x} - \sqrt{\sin x})(\sqrt{\cos x} + \sqrt{\sin x})}{3(\cos x + \sin x)} \right $ | (Q) | $\left[\frac{4}{3}, \frac{7}{3}\right]$ |
| (III) | $\frac{7}{3(x^6 + 2x^4 + 3x^2 + 1)}$ | (R) | $\left[0, \frac{1}{3}\right]$ |
| (IV) | $\log_8(x^2 + 2x + 2)$ | (S) | $[0, \infty)$ |
- Which of the following is correct ?
 (A) $I \rightarrow Q; II \rightarrow R; III \rightarrow P; IV \rightarrow S$ (B) $I \rightarrow Q; II \rightarrow R; III \rightarrow S; IV \rightarrow P$
 (C) $I \rightarrow P; II \rightarrow S; III \rightarrow Q; IV \rightarrow R$ (D) $I \rightarrow P; II \rightarrow S; III \rightarrow R; IV \rightarrow Q$

MFN077

MATRIX MATCH TYPE QUESTION

- | 19. | Column-I | | Column-II |
|-----|--|-----|-------------|
| | Number of integers in | | |
| (A) | Domain of $f(x) = \ln\{x\}$ | (P) | 0 |
| (B) | Domain of $f(x) = \sqrt{\sec(\sin x)} + \sqrt{\left[x + \frac{1}{x}\right]} + \sqrt{10 - [x]^2}$ | (Q) | 2 |
| (C) | Range of $f(x) = x^2 - 2x + 2, x \in [0, 2]$ | (R) | 3 |
| (D) | Range of $f(x) = \sqrt{25 - [x]^2}$ | (S) | less than 3 |
| | | (T) | more than 3 |

(where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively)

MFN078

EXERCISE (S)

1. The sum of integral values of the elements in the domain of $f(x) = \sqrt{\log_{\frac{1}{2}}|3-x|}$ is
MFN021
2. Let $f(x) = \frac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$
MFN022
3. The number of integral values of x satisfying the inequality $[x-5][x-3] + 2 < [x-5] + 2[x-3]$ (where $[.]$ represents greatest integer function) is -
MFN023
4. Suppose, $f(x, n) = \sum_{k=1}^n \log_x \left(\frac{k}{x}\right)$, then the value of x satisfying the equation $f(x, 10) = f(x, 11)$, is
MFN024
5. If $f(x) = a \log \left(\frac{1+x}{1-x}\right) + bx^3 + c \sin x + 5$ and $f(\log_3 2) = 4$, then $f\left(\log_3 \left(\frac{1}{2}\right)\right)$ is equal to
MFN025
6. If $f(x) = px + q$ and $f(f(f(x))) = 8x + 21$, where p and q are real numbers, then $p + q$ equals-
MFN026
7. Let $f(x) = \frac{2x-1}{x+3}$. If $f^{-1} = \frac{ax+b}{c-x}$, then $a + b + c$ is
MFN027
8. A function ' f ' is defined for all real numbers and satisfies $f(2+x) = f(2-x)$ and $f(7+x) = f(7-x)$ for all real x . If $x = 0$ is a root of $f(x) = 0$, then find least number of roots of $f(x) = 0$ for $x \in [-1000, 1000]$.
MFN028
9. If $f(x, y) = \max(x, y) + \min(x, y)$ and $g(x, y) = \max(x, y) - \min(x, y)$, then the value of $f\left(g\left(-\frac{2}{3}, -\frac{3}{2}\right), g(-3, -4)\right)$ is equal to $\frac{k}{6}$ then k ($k \in \mathbb{N}$) is -
MFN029
10. Let ' f ' be an even periodic function with period '4' such that $f(x) = 2^x - 1$, $0 \leq x \leq 2$. The number of solutions of the equation $f(x) = 1$ in $[-10, 20]$ are
MFN030

EXERCISE - JEE (Main) PYQ

1. For $x \in R - \{0, 1\}$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2 \circ f_1)(x) = f_3(x)$ then $J(x)$ is equal to :- [JEE (Main) 2019]
 (1) $f_3(x)$ (2) $f_1(x)$ (3) $f_2(x)$ (4) $\frac{1}{x} f_3(x)$
MFN031
2. If $f(x) = \log_e \left(\frac{1-x}{1+x} \right)$, $|x| < 1$, then $f\left(\frac{2x}{1+x^2}\right)$ is equal to : [JEE (Main) 2019]
 (1) $2f(x)$ (2) $2f(x^2)$ (3) $(f(x))^2$ (4) $-2f(x)$
MFN035
3. Let $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$, where the function f satisfies $f(x+y) = f(x)f(y)$ for all natural numbers x, y and $f(1) = 2$. then the natural number 'a' is [JEE (Main) 2019]
 (1) 4 (2) 3 (3) 16 (4) 2
MFN036
4. If the function $f : R - \{1, -1\} \rightarrow A$ defined by $f(x) = \frac{x^2}{1-x^2}$, is surjective, then A is equal to [JEE (Main) 2019]
 (1) $R - [-1, 0)$ (2) $R - (-1, 0)$ (3) $R - \{-1\}$ (4) $[0, \infty)$
MFN037
5. The domain of the definition of the function $f(x) = \frac{1}{4-x^2} + \log_{10}(x^3 - x)$ is :- [JEE (Main) 2019]
 (1) $(1, 2) \cup (2, \infty)$ (2) $(-1, 0) \cup (1, 2) \cup (3, \infty)$
 (3) $(-1, 0) \cup (1, 2) \cup (2, \infty)$ (4) $(-2, -1) \cup (-1, 0) \cup (2, \infty)$
MFN038
6. If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to [JEE (Main) 2020]
 (1) $\frac{3}{2}$ (2) $-\frac{1}{2}$ (3) $-\frac{3}{2}$ (4) $\frac{1}{2}$
MFN039
7. Let $f : (1, 3) \rightarrow R$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is [JEE (Main) 2020]
 (1) $\left(\frac{3}{5}, \frac{4}{5}\right)$ (2) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$ (3) $\left(\frac{2}{5}, \frac{4}{5}\right]$ (4) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$
MFN040
8. Let $f : R \rightarrow R$ be such that for all $x \in R$ $(2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is [JEE (Main) 2020]
 (1) 0 (2) 3 (3) 2 (4) 4
MFN041

9. The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}, x \in (-1, 1)$, is [JEE (Main) 2020]
- (1) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1-x}{1+x} \right)$ (2) $\frac{1}{4} \log_e \left(\frac{1-x}{1+x} \right)$
 (3) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1+x}{1-x} \right)$ (4) $\frac{1}{4} \log_e \left(\frac{1+x}{1-x} \right)$
- MFN042
10. Let $f: R \rightarrow R$ be a function which satisfies $f(x+y) = f(x) + f(y) \forall x, y \in R$. If $f(1) = 2$ and $g(n) = \sum_{k=1}^{(n-1)} f(k), n \in N$ then the value of n , for which $g(n) = 20$, is : [JEE (Main) 2020]
- (1) 5 (2) 9 (3) 20 (4) 4
- MFN043
11. Let $[t]$ denote the greatest integer $\leq t$. Then the equation in $x, [x]^2 + 2[x+2] - 7 = 0$ has : [JEE (Main) 2020]
- (1) no integral solution (2) exactly four integral solutions
 (3) exactly two solutions (4) infinitely many solutions
- MFN044
12. For a suitably chosen real constant a , let a function, $f: R - \{-a\} \rightarrow R$ be defined by $f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a, (f \circ f)(x) = x$. Then $f\left(-\frac{1}{2}\right)$ is equal to : [JEE (Main) 2020]
- (1) $\frac{1}{3}$ (2) 3 (3) -3 (4) $-\frac{1}{3}$
- MFN045
13. Suppose that a function $f: R \rightarrow R$ satisfies $f(x+y) = f(x)f(y)$ for all $x, y \in R$ and $f(1) = 3$. If $\sum_{i=1}^n f(i) = 363$, then n is equal to. [JEE (Main) 2020]
- MFN046
14. The range of the function, $f(x) = \log_{\sqrt{5}} \left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right) \right)$ is : [JEE (Main) 2021]
- (1) $(0, \sqrt{5})$ (2) $[-2, 2]$ (3) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (4) $[0, 2]$
- MFN047
15. Let $f(x)$ be a polynomial of degree 3 such that $f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10 f(10)$ is equal to : [JEE (Main) 2021]
- MFN048
16. The inverse of $y = 5^{\log x}$ is : [JEE (Main) 2021]
- (1) $x = 5^{\log y}$ (2) $x = y^{\log 5}$ (3) $x = y^{\frac{1}{\log 5}}$ (4) $x = 5^{\frac{1}{\log y}}$

MFN049

17. Let $f: R - \{3\} \rightarrow R - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g: R \rightarrow R$ be given as $g(x) = 2x - 3$. Then,

the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to **[JEE (Main) 2021]**

- (1) 7 (2) 2 (3) 5 (4) 3

MFN050

18. Let $[x]$ denote the greatest integer $\leq x$, where $x \in R$. If the domain of the real valued function

$f(x) = \sqrt{\frac{[x]-2}{[x]-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty)$, $a < b < c$, then the value of $a + b + c$ is:

[JEE (Main) 2021]

- (1) 8 (2) 1 (3) -2 (4) -3

MFN051

19. The number of bijective functions $f: \{1, 3, 5, 7, \dots, 99\} \rightarrow \{2, 4, 6, 8, \dots, 100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \dots \geq f(99)$, is. **[JEE (Main) 2022]**

- (1) ${}^{50}P_{17}$ (2) ${}^{50}P_{33}$ (3) $33! \times 17!$ (4) $\frac{50!}{2}$

MFN052

20. Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p$, $p \neq 0$ and $f(1) = \frac{1}{3}$. If the equation $f(x) = 0$ and $f(f(f(f(f(x)))) = 0$ have a common real root, then $f(-3)$ is equal to. **[JEE (Main) 2022]**

MFN053

21. The total number of functions, $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to : **[JEE (Main) 2022]**

- (1) 60 (2) 90 (3) 108 (4) 126

MFN054

22. Let $f: R \rightarrow R$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal

to : **[JEE (Main) 2022]**

- (1) 4 (2) 10 (3) 11 (4) 16

MFN055

23. Let $f: R \rightarrow R$ be defined as $f(x) = x - 1$ and $g: R - \{1, -1\} \rightarrow R$ be defined as $g(x) = \frac{x^2}{x^2 - 1}$.

Then the function fog is :

[JEE (Main) 2022]

- (1) one-one but not onto function (2) onto but not one-one function
(3) both one-one and onto function (4) neither one-one nor onto function

MFN056

24. The number of one-one function $f: \{a, b, c, d\} \rightarrow \{0, 1, 2, \dots, 10\}$ such that

$2f(a) - f(b) + 3f(c) + f(d) = 0$ is.

[JEE (Main) 2022]

MFN057

25. Let $f: N \rightarrow R$ be a function such that $f(x+y) = 2f(x)f(y)$ for natural numbers x and y . If $f(1) = 2$, then the value of α for which $\sum_{k=1}^{10} f(\alpha+k) = \frac{512}{3}(2^{20}-1)$ holds, is [JEE (Main) 2022]
 (1) 2 (2) 3 (3) 4 (4) 6 MFN058
26. Let $f: R \rightarrow R$ be a function defined by $f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2+x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, the greatest integer less than or equal to $g(1)$ is [JEE (Main) 2022]
 MFN059
27. If domain of the function $\log_e \left(\frac{6x^2+5x+1}{2x-1} \right) + \cos^{-1} \left(\frac{2x^2-3x+4}{3x-5} \right)$ is $(\alpha, \beta) \cup (\gamma, \delta]$, then $18(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$ is equal to. [JEE (Main) 2023]
 MFN088
28. Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto function $f: R \rightarrow S$ such that $f(a) \neq 1$, is equal to _____. [JEE (Main) 2023]
 MFN089
29. Let $f: R - \{0, 1\} \rightarrow R$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1+x$. Then $f(2)$ is equal to : [JEE (Main) 2023]
 (1) $\frac{9}{2}$ (2) $\frac{9}{4}$ (3) $\frac{7}{4}$ (4) $\frac{7}{3}$ MFN090
30. Consider a function $f: \mathbb{N} \rightarrow \mathbb{R}$, satisfying $f(1) + 2f(2) + 3f(3) + \dots + xf(x) = x(x+1)f(x); x \geq 2$ with $f(1)=1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to [JEE (Main) 2023]
 (1) 8200 (2) 8000 (3) 8400 (4) 8100 MFN091
31. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of $f, \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha+1}$ is equal to [JEE (Main) 2024]
 MFN141
32. Let $f, g: R \rightarrow R$ be defined as : $f(x) = |x-1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$. Then the function $f(g(x))$ is [JEE (Main) 2024]
 (1) neither one-one nor onto. (2) one-one but not onto.
 (3) both one-one and onto. (4) onto but not one-one. MFN142

Functions

33. If $S = \{a \in R : |2a - 1| = 3[a] + 2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t , then $72 \sum_{a \in S} a$ is equal to _____. **[JEE (Main) 2024]**

MFN143

34. Let $[t]$ be the greatest integer less than or equal to t . Let A be the set of all prime factors of 2310 and $f : A \rightarrow \mathbb{Z}$ be the function $f(x) = \left\lfloor \log_2 \left(x^2 + \left\lceil \frac{x^3}{5} \right\rceil \right) \right\rfloor$. The number of one-to-one functions from A

to the range of f is :

[JEE (Main) 2024]

- (1) 20 (2) 120 (3) 25 (4) 24

MFN144

35. The function $f : N - \{1\} \rightarrow N$; defined by $f(n)$ = the highest prime factor of n , is :

[JEE (Main) 2024]

- (1) both one-one and onto (2) one-one only
(3) onto only (4) neither one-one nor onto

MFN145

EXERCISE - JEE (Advanced) PYQ

1. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is - [IIT JEE 2011]
 (A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$
 (B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
 (C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
 (D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
MFN080
2. The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is : [IIT JEE 2012]
 (A) one-one and onto
 (B) onto but not one-one
 (C) one-one but not onto
 (D) neither one-one nor onto
MFN081
3. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then:- [JEE (Advanced) 2014]
 (A) $f(x)$ is an odd function
 (B) $f(x)$ is a one-one function
 (C) $f(x)$ is an onto function
 (D) $f(x)$ is an even function
MFN083
4. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one one functions from X to Y and β is the number of onto functions from Y to X , then the value of $\frac{1}{5!}(\beta - \alpha)$ is_____ [JEE (Advanced) 2018]
MFN084
5. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE? [JEE (Advanced) 2020]
 (A) f is one-one, but NOT onto
 (B) f is onto, but NOT one-one
 (C) f is BOTH one-one and onto
 (D) f is NEITHER one-one NOR onto
MFN085
6. Let the function $f: [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$
 Then the value of $f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \dots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$ is_____ [JEE (Advanced) 2020]
MFN086

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x^2 \sin\left(\frac{\pi}{x^2}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Then which of the following statements is TRUE ?

[JEE (Advanced) 2024]

(A) $f(x) = 0$ has infinitely many solutions in the interval $\left[\frac{1}{10^{10}}, \infty\right)$.

(B) $f(x) = 0$ has no solutions in the interval $\left[\frac{1}{\pi}, \infty\right)$.

(C) The set of solutions of $f(x) = 0$ in the interval $\left(0, \frac{1}{10^{10}}\right)$ is finite.

(D) $f(x) = 0$ has more than 25 solutions in the interval $\left(\frac{1}{\pi^2}, \frac{1}{\pi}\right)$.

MFN146

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, and $g : \mathbb{R} \rightarrow (0, \infty)$

be a function such that $g(x + y) = g(x)g(y)$ for all $x, y \in \mathbb{R}$. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then

the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is _____.

[JEE (Advanced) 2024]

MFN147

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	B	B	D	A	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	A	D	C	C	A	D	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,C	A,B,C	B,C,D	C,D	A,B,C,D	A,B,C	A,D	B,C	A,D
Que.	11	12	13	14	15	16	17	18		
Ans.	A,B,C	A	C	A	D	D	C	A		
Que.	19									
Ans.	$A \rightarrow P,S ; B \rightarrow R ; C \rightarrow Q,S ; D \rightarrow T$									

EXERCISE (S)

1.	6	2.	1002.50	3.	2	4.	11	5.	6
6.	5	7.	6	8.	401	9.	11	10.	15

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	1	3	2	4	2	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	5.00	4	26	3	3	3	2	25
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	4	31	3	2	20	180	2	4
Que.	31	32	33	34	35					
Ans.	1024	1	18	2	4					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8
Ans.	A	B	A,B,C	119	C	19.00	D	51